Question	Answer	Marks
1	**Defining the problem**	
	vary *f* and measure *V* **or** *f* is the independent variable and *V* is the dependent variable	**1**
	keep *E* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • circuit with a.c. supply • oscilloscope connected in parallel with the resistor • workable circuit • oscilloscope and a.c. supply labelled	**1**
	labelled signal generator or variable frequency power supply connected across the terminals	**1**
	method to determine *V* or *E* from oscilloscope, e.g. multiply amplitude / height of wave by *y*-gain on oscilloscope	**1**
	method to determine *f* from oscilloscope, e.g. determine period *T* by multiplying number of divisions in 1 cycle or horizontal distance in 1 cycle by the time base **and** $f = 1/T$	**1**
	Method of Analysis	
	plot a graph of $\frac{1}{V}$ against *f* or equivalent, e.g. *f* against $\frac{1}{V}$ Allow logarithms e.g. lg *V* against lg *f*.	**1**
	relationship valid if a straight line is produced passing through the origin (for lg *V* against lg *f*: relationship valid if a straight line is produced with gradient = −1)	**1**

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Question	Answer	Marks
1	<table><tr><td>$\frac{1}{V}$ against f</td><td>f against $\frac{1}{V}$</td></tr><tr><td>$K = \frac{lES}{AN^2} \times$ gradient</td><td>$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$</td></tr></table> (for lg V against lg f: $K = \frac{ElS}{AN^2} \times 10^{-y\text{-intercept}}$).	**1**
	Additional detail including safety considerations	**6**
	D1 precaution linked to hot coil or hot resistor or prevention of burns from coil or resistor, e.g. use gloves / switch off power supply when not measuring V to prevent burns from coil / resistor	
	D2 keep N **and** A **and** l **and** S constant	
	D3 method to keep S constant, e.g. switch off power supply between readings to prevent heating of resistor or to allow resistor to cool	
	D4 method to determine A, e.g. use calipers / micrometer to measure diameter (of coil) / d **and** $A = \frac{\pi d^2}{4}$	
	D5 repeat measurements of diameter d along the length of the coil / in different directions **and** determine the average value of d	
	D6 method to determine the value of S, e.g. separate circuit diagram showing resistor connected to ohmmeter, or circuit diagram showing resistor connected to a power supply with an ammeter and voltmeter **and** $S = V / I$	
	D7 measure l with a ruler / calipers	
	D8 oscilloscope drawn connected across terminals / across signal generator **and** description to determine E	
	D9 adjust y-gain for maximum amplitude **or** adjust time base for length of one wave or measure n waves and divide measured time by n	

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Question	Answer	Marks
1	D10 method to keep *E* constant, e.g. check p.d. and alter supply **or** method to keep l constant, e.g. tape coil **or** method to keep *A* constant, e.g. wind wire on a cylinder	

Question	Answer	Marks
2(a)	gradient = $\frac{R}{E}$ *y*-intercept = $\frac{Z}{E}$	**1**
2(b)	$\frac{1}{I}$ / 10^3 A^{-1} 2.20 or 2.198 1.90 or 1.905 1.72 or 1.724 1.57 or 1.575 1.46 or 1.460 1.31 or 1.307 Values of $\frac{1}{I}$ / 10^3 A^{-1} correct as shown above.	**1**

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